Company Primer

CuspAI

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Contributing Experts

FORMERS

- **CuspAI** - Former Specialist

COMPETITORS

- **Alice (Formerly ActiveFence)** - Specialist
- **Aionics** - Executive
- **SandboxAQ** - Vice-President
- **Citrine Informatics** - Executive
- **Citrine Informatics** - Executive

CUSTOMERS

- **Borouge** - Head
- **OTI Lumionics** - Vice-President
- **Medtronic** - Director

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Overview

CuspAI uses GenAI to discover new materials for semiconductor & electronics, MedTech & pharma, and chemical & energy companies. The company formulates and validates catalysts, sorbents, and electronic materials against defined material property targets. These materials are used for carbon capture, chip cooling, and water remediation.

Founded	2024
HQ	Cambridge
Revenue	£7m to £12m (2025) ↑ 100% to 150% p.a. (2025)
FTEs	51 to 200 (2026)
Ownership	VC backed (Series B)
Markets	AI-Powered Materials Discovery - Primary

MARKET CONTEXT

PARTICIPANTS

AI-Powered Materials Discovery Providers	Companies such as CuspAI, Orbital Industries, and Citrine Informatics develop AI models and platforms that discover and validate new materials. Providers also offer end-to-end R&D projects and support.
Customers	These include chemical, semiconductor, energy, and MedTech companies that contract providers to discover and optimize materials. KDMs are often R&D directors, and users include material scientists and engineers.
Partners	CuspAI has a global network of partners through its 'AI Materials Foundry', including companies like 3M, Meta, Samsung, NVIDIA, and ASML. Partners co-develop models with the company and help it calibrate its capabilities to real-world applications. Cloud and chip partners like NVIDIA also supply compute capacity to providers. NVIDIA invests in many providers and lowers compute costs for its partners.
Government and Academic Institutions	Government bodies and universities fund and purchase material research. Government grants drive higher demand for CuspAI by supporting customers' budgets for AI spend, and public AI initiatives make institutions a potential customer segment.
Regulators	Environmental regulations define rules on carbon emissions, and the use of per- and polyfluoroalkyl substances (PFAS), driving material discovery demand. Chemical regulators oversee material compliance and safety controls, increasing R&D cycle frequency.

OVERVIEW

Market Need: chemical, semiconductor, and energy companies compete to discover new materials that can improve their product and margin performance. These customers invest in identifying new catalysts, polymers, coatings, and electronic materials to drive differentiation and profitability within their industries requiring agile R&D. Material discovery through traditional methods often takes years of laboratory iterations, and internal R&D teams can test only a limited number of candidates. AI-powered materials discovery providers generate and simulate candidate materials computationally, enabling customers to screen more candidates in weeks and cut material screening time.

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Before these tools, customers relied on in-house computational chemistry teams running DFT and molecular dynamics simulations, supplier-led material selection, and manual laboratory trial and error. Adoption of AI tools has grown since 2023 as new products entered the market covering more applications, and customer confidence has grown as early projects delivered successful ROI results. In-house teams remain the largest substitute, since full-stack chemical producers can build internal capability at a lower cost than external providers charge.

Market Structure: The TAM could reach £25bn to £40bn including laboratory testing and system integration if every chemical and semiconductor company adopted AI-powered discovery tools across R&D departments.

However, penetration in 2026 remains low, limited by a lack of in-house computational talent among customers and confidentiality concerns.

- Approximately 2% of R&D budgets of addressable companies are allocated to external material discovery providers.
- Approximately 50% of addressable customers use AI-powered material discovery tools in their R&D processes, including in-house development of these capabilities.
- Approximately 30% of addressable customers purchase AI-powered material discovery tools from external providers.

The AI-powered materials discovery market generated £200m to £250m in revenue in 2025. Early demand concentrates in chemicals, semiconductors, batteries, and sustainability applications like carbon capture and PFAS removal.

Growth: market revenue is growing at 10% to 15% annually and is expected to maintain this rate until 2030, driven by productivity gains, add-on capabilities, growing customer confidence, and sustainability compliance requirements. Startups like CuspAI are growing faster than the market at 50% to 100% annually, since most materials companies have yet to adopt external AI tool providers, and large whitespace drives efficient customer acquisition.

SERVICES AND PRODUCTS

Custom Materials Discovery Programs	This is an end-to-end R&D engagement in which CuspAI teams discover and optimize materials against target properties. CuspAI's Program Directors run testing, simulations, and material candidate selection rather than the customer.
MIRA Platform Access and Simulation	This includes access to CuspAI's in-house GenAI models and simulation tools. Customers run searches that generate candidate materials and simulate candidate behavior before laboratory testing. It also includes model guidance and technical support.

EXAMPLE USE CASE - Specialty Chemicals Customer Discovering a Catalyst Through a Custom Program

A specialty chemicals manufacturer with £3bn in revenue wants a new catalyst, a material that speeds up chemical reactions in fuel refining, that raises its output by 5% compared to the product it uses today. The VP of R&D owns the budget and hires CuspAI after a six-month evaluation of competing providers, and the two organizations agree on what the new material must achieve, the project milestones, and a fixed project fee with payments tied to each stage.

CuspAI's scientists convert the performance requirements into instructions for its AI models, and the models generate thousands of candidate materials on a computer. Instead of testing each candidate in a laboratory, which would take years, the team uses physics-based simulations to predict how each candidate would behave, and the platform ranks the candidates against the customer's requirements. Catalyst researchers at the customer review the shortlists each week and adjust the search, and laboratory technicians test the most promising candidates in pilot reactors.

Within eight months, the program identifies three candidates that outperform the customer's existing catalyst in simulation. CuspAI's validation partners produce small physical samples of each candidate, and the customer tests them under real refinery conditions. The manufacturer scales the winning candidate into full production and pays CuspAI a royalty on the materials it produces with the discovery. The customer's Director of R&D uses the results to justify expanding the program to a second catalyst family.

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NOTABLE EVENTS

2024, Q1 — **Founded**

CuspAI was founded by Dr. Chad Edwards, a former Google DeepMind chemist, and Prof. Max Welling, a former scientist at Microsoft Research. It was founded as a generative materials discovery startup, entering a market where enterprise informatics providers had operated for more than a decade.

2024, Q2 — **Hoxton Ventures led CuspAI's seed round**

The company raised £20m to £25m through its seed funding round. Basis Set Ventures and Lightspeed Venture Partners participated in this funding round, which enabled initial model development.

2025, Q3 — **NEA led CuspAI's Series A funding round**

It raised more than £75m through its Series A funding at a post-money valuation of approximately £400m. Bezos Expeditions and the UK Sovereign AI Fund participated in this funding round, which enabled it to improve internal capabilities to serve customers.

2026, Q3 — **Kleiner Perkins led CuspAI's Series B funding round**

CuspAI raised £300m to £350m through its Series B funding at a valuation of approximately £2bn. Bezos Expeditions and the UK Sovereign AI Fund participated in this funding round, which enables further model development and customer acquisition as it scales custom program delivery.

2026, Q3 — **CuspAI launched its 'AI Materials Foundry' program**

This is a proprietary ecosystem of more than 45 enterprise and lab partners that collaborate on CuspAI's platform. This enabled the company to improve its credibility in the market.

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SEGMENTATION

All revenue splits are illustrative. They are taken from expert estimates and corroborated with public sources where available.

CUSTOMERS

SEGMENTS	SPLIT (2025)
Semiconductors & Electronics	45%-55%
Chemicals, Energy, & Others	40%-50%
MedTech & Pharma	0%-10%

Semiconductors & Electronics Customers

TREND ↑ Increasing

COMMENTARY

Customer Profile: this segment includes chipmakers and electronic and semiconductor equipment producers that engage CuspAI to discover materials applied around the chip. Customers use CuspAI's offerings to discover photoresists, polishing compounds, packaging materials, and thermal interface materials to manage high heat loads in high-performance computing and AI hardware.

Demand Drivers:

Customers in this segment do not attempt to replace silicon as it is well-understood and a better substitute is less likely to be manufactured at scale, limiting CuspAI's applications to this segment. However, they face sustained competitive pressure to innovate quickly as the sector is growing and chip buyers prioritize the most efficient products over the less costly. This makes chipmakers invest in R&D efforts around innovative materials to support chip performance required to remain competitive. CuspAI covers chip-optimizing applications through its MIRA platform, enabling it to capture this demand.

Semiconductors & Electronics customers' R&D budgets are growing quickly as increased demand for chips is driving increased earnings. This makes customers in this segment less price-sensitive than chemical buyers, enabling CuspAI to charge higher prices.

ACV Dynamics: semiconductor customers allocated less than 1% of R&D budgets to AI-powered materials discovery in 2025 due to strict system integration requirements. Every material is tightly integrated into manufacturing workflows, limiting how many new materials each customer can absorb in a year. This drives lower ACVs in this segment as annual testing volume and platform usage are limited by integration feasibility.

Growth: CuspAI is expected to grow revenue at a CAGR of 40% to 60% until 2029 in this segment, driven by increased adoption of AI-powered material discovery and new customer acquisition. The share of customers' R&D spend allocated to providers is expected to reach approximately 5% by 2029 as platforms demonstrate initial ROI in specific use cases and customers increase spend. Further growth could be limited by the integrated structure of semiconductor manufacturing.

Chemicals, Energy, & Others Customers

TREND ↑ Increasing

EXAMPLE CUSTOMERS

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COMMENTARY

Customer Profile: this segment includes specialty chemicals producers, battery manufacturers, oil & gas (O&G), and clean energy companies. Customers typically use CuspAI to discover and test catalysts, sorbents, PFAS-free materials, carbon capture materials, and battery chemistries.

Demand Drivers:

Customers in this segment typically have low EBITDA margins, driving spend in R&D projects generating incremental performance gains. Demand is driven by national COP commitments, like the UAE 2030 and 2045 ESG goals, and strict environmental regulations covering sulfur levels and PFAS elimination. These commitments push customers to discover compliant replacement materials and increase discovery budgets according to public sustainability targets.

Chemicals, energy, & other customers allocate 1% to 2% of R&D spend to AI-powered material discovery providers. In-house R&D and material discovery are the largest substitutes for platforms like CuspAI in this segment as the largest chemical producers can build comparable capabilities internally at a lower cost than external providers charge.

ACV and Product Mix Dynamics:

ACVs in this segment are up to 2x higher than for semiconductors & electronics due to multiple use cases across customers' R&D portfolios. The MIRA platform is used for refining, carbon management, environmental remediation, corrosion-resistant steels, and other applications, driving increased platform usage in this customer segment.

Chemical customers typically purchase platform access and simulation rather than custom R&D programs due to IP concerns and royalties. Proprietary molecules and formulation datasets are core value drivers for these customers, making them hesitant to share this information with external providers through custom R&D engagements. Custom programs also typically charge higher royalties on discovered materials, limiting customer profitability when they create value from these activities.

Growth: CuspAI is expected to grow revenue at a CAGR of 40% to 60% until 2029 in this segment, driven by wider application coverage and increased spend in existing accounts. CuspAI is driving increased spending as customers move from pilot stages to long-term contracts including integrated AI workflows and more use cases.

The percentage of R&D budgets spent by customers in external material discovery providers is expected to reach 5% to 10% by 2029 as successful pilots and improving capabilities drive adoption. Battery and clean energy customers could drive further growth as government and private investment returns to those categories and increases R&D budgets.

MedTech & Pharma Customers

TREND ↓ Decreasing

EXAMPLE CUSTOMERS Medtronic

COMMENTARY

Customer Profile: this segment includes medical device manufacturers and pharmaceutical companies that engage CuspAI to discover polymers, coatings, and adhesives for regulated products.

Demand Drivers:

Device manufacturers must test material properties like flexibility, durability, adhesion, and biocompatibility for every new product development. CuspAI enables customers in this segment to screen and test risks related to candidate materials earlier than conventional laboratory testing, which could reveal risks after months or years of development. This enables customers to achieve shorter development cycles and faster time to market.

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Pharmaceutical companies allocate 2% to 3% of R&D budgets to AI-driven material discovery platforms, as discovery outcomes have larger financial upside. However, this spend is concentrated in drug discovery, which is not CuspAI's specialization and target market. This drives the majority of demand towards competitors that provide models focused on drug discovery like Lila Sciences.

ACV Dynamics: pharmaceutical customers typically have larger ACVs as drug discovery can create revenues above £1bn for these customers. Medtech customers have smaller ACVs than pharmaceutical buyers and run parallel pilots with several providers before selection. This drives lower ACVs for CuspAI in this segment compared to semiconductors & electronics.

Growth: CuspAI is expected to grow at a CAGR of 20% to 30% until 2029 in this segment as customer adoption grows. The company directs model development towards industrial chemistries, maintaining growth in this segment concentrated in OEMs attempting to discover and optimize polymers and soft materials. It could grow faster if it develops dedicated models to target pharmaceutical applications, given this segment requires dedicated chemistry expertise.

PRODUCTS

SEGMENTS	SPLIT (2025)
Custom Materials Discovery Programs	55%-65%
MIRA Platform Access and Simulation	35%-45%

Custom Materials Discovery Programs

Products

TREND ↑ Increasing

EXAMPLE CUSTOMERS

COMMENTARY

Customers typically require custom programs rather than basic platform access. This includes co-development projects, consulting services, and proprietary model tuning. CuspAI scientists optimize customer usage of its models by facilitating integration and tailoring models to customers' requirements.

Demand Drivers: customers typically lack in-house AI-driven R&D capabilities and computational chemistry expertise to use AI discovery tools without support. Customers with internal teams typically focus on specific material families, requiring CuspAI's services for specialized materials. The novelty and complexity of AI-powered material discovery also drive customers to require CuspAI staff to guide usage for the initial months before basic platform subscriptions can be effective.

ACV Dynamics: Custom programs have higher contract values than platform access as CuspAI provides dedicated scientific staff for each project. ACVs are £100k to £1m in this segment in 2025 as customers test custom programs across a limited set of materials before committing to a larger investment and full integration.

Growth: CuspAI is expected to grow at more than 100% in 2026 in this segment, driven by new customer acquisition and incremental spend of existing customers. It is expected to grow at a CAGR of around 40% between 2027 and 2029 in this segment as the revenue base expands. Custom programs and consulting are expected to account for 60% to 70% of incremental spending by existing customers in 2026 and 2027. From 2026 to 2029, it could generate ACVs up to £4m for large industrial and petrochemical customers requiring a wide material coverage as they transition from pilots to complete, multi-year engagements. New use cases and material families require additional dedicated scientific support.

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MIRA Platform Access and Simulation Products

TREND ↑ Increasing

EXAMPLE CUSTOMERS Medtronic

COMMENTARY

This product includes access to CuspAI's models and simulation tools through a subscription which is typically bundled with consulting services like model guidance and human-led technical support.

Value Proposition and Demand:

The MIRA platform enables customers to reduce material screening time by 20% to 25% and trial-and-error laboratory iterations by approximately 15%. Faster screening shortens the length of R&D projects and facilitates faster applications of new materials. This enables customers to remain competitive in product launches where short time to market is required, driving higher demand for CuspAI.

CuspAI's model uses physics-based simulations to model structure-property relationships, enabling customers' R&D teams to validate inputs, outputs, and property constraints. The platform also has a high external audit defensibility due to its explainable and transparent modeling rather than the probabilistic text generation provided by general AI assistants like ChatGPT and Perplexity. Customers in regulated industries require validated and explainable models to withstand auditor scrutiny, driving demand for specialized discovery platforms and retention.

Customers who prefer process IP confidentiality often prefer platform access over custom programs. Internal platform operation enables customers to maintain process knowledge in-house rather than sharing R&D processes and expertise with external consultants. Basic platform access enables CuspAI to serve customers that prefer to retain proprietary research information within their own organizations.

ACV Dynamics: Platform access has contract values more than 50% smaller than custom programs due to lower service levels. This also drives lower cost to serve customers in this segment and enables CuspAI to increase its margins.

Growth: CuspAI is expected to grow at 20% to 30% in 2026 in this segment as new customers must purchase platform access to engage in custom programs. It is expected to grow at a CAGR of 10% to 15% in this segment between 2027 and 2028 as it focuses on advisory and co-development engagements. The software platform has not reached the maturity level where most customers are able to operate it without using consulting services like human-led scientific support given continuous changes in the technology.

REGIONS

SEGMENTS	SPLIT (2025)
EMEA	50%-60%
Americas	30%-40%
APAC	5%-15%

EMEA Regions

TREND ↓ Decreasing

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EXAMPLE CUSTOMERS

COMMENTARY

CuspAI has its largest presence in Europe, where revenue generation concentrates in chemical and industrial companies requiring services for PFAS removal, carbon capture, and catalyst applications. Customers in this region require a wider material coverage to meet more stringent compliance standards, driving increased spend in CuspAI.

Customers operating in countries in the Gulf Cooperation Council (GCC) require on-the-ground support and data residency agreements due to national regulations and sovereign data localization requirements, increasing relative costs to serve. Citrine Informatics provides system integration of material discovery solutions beyond CuspAI's core R&D department capabilities. This drives lower win rates for CuspAI in this region as customers prioritize data infrastructure and system integration.

Regulation: European regulation on PFAS restrictions and environmental toxins is expected to become stricter, driving increased demand from chemical, energy, & other petrochemical companies for compliant replacement materials. It also supports further customer acquisition as these regulations incentivize more companies to pursue material discovery and expand CuspAI's addressable market.

Growth: CuspAI is expected to grow at 25% to 30% annually in this region between 2026 and 2029, driven by regulatory scrutiny and increased material coverage from customers. Growth is limited by high input costs and limited cash flow facing European chemical companies, which decreases their ability to invest in new discovery programs. Energy input disruptions due to geopolitical conflicts are raising costs for European producers more than in the Americas, reducing the budgets available for AI adoption.

Americas

Regions

TREND ↑ Increasing

EXAMPLE CUSTOMERS

COMMENTARY

Penetration Dynamics: US companies operate with less centralized decision-making than APAC companies, enabling providers to transact with individual business units and run smaller test deals before enterprise commitments. This facilitates CuspAI's penetration into customers with high spending potential to run pilot projects and convert them into long-term contracts.

Adoption: US customers have adopted AI-driven material discovery tools more slowly than in Europe and Asia, leaving larger whitespace for CuspAI to capture. Competitive pressures are expected to drive adoption between 2026 and 2029 among chemicals and energy companies in the Americas.

Government Demand: government funding directed to technology and materials companies raises the budgets available for AI discovery engagements. The US government has AI incentive programs that enable semiconductor manufacturers to fund projects for chip packaging, photoresists, and cooling liquids optimization. North American governments are also increasing internal budgets for AI-powered solutions that support sovereignty and reduce reliance on imports of novel materials. This would make governments a large customer segment with high growth potential.

Growth: CuspAI is expected to grow at a CAGR of 40% to 50% in this region between 2026 and 2029, driven by rising awareness of competitive pressure from Chinese AI investment, and larger R&D budgets. US chemical producers benefit from favorable access to North American feedstocks, which protects cash flow and leaves larger budgets available for discovery programs than European producers. High AI adoption from China is expected to support continuous US government incentives for semiconductor and chemical & energy companies to adopt AI solutions like CuspAI's.

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APAC

Regions

TREND

↑ Increasing

COMMENTARY

Korea and Japan represent the largest opportunity in APAC as CuspAI does not operate in China. The Chinese government prioritizes domestic companies, which drive large-scale AI-driven materials research. Battery and electronics producers in Korea and Japan like LG, Samsung, and Panasonic face competitive pressure from Chinese manufacturers like CATL that use government-backed AI-driven material discovery providers on a large scale. This creates demand in these countries for non-Chinese material discovery solutions like CuspAI's to support electronics and battery innovation.

APAC companies typically operate under more centralized decision-making, limiting penetration in this region for pilot engagements. Procurement and regulatory vetting often rely on a central committee, requiring enterprise-wide approval and negotiation rather than individual BUs. This lengthens sales cycles and limits conversion of leads, but drives higher ACVs after successful penetration as wider adoption across the company is facilitated.

Growth: CuspAI is expected to grow at a CAGR of 60% to 75% in APAC between 2026 and 2029, driven by large whitespace and competitive pressure on customers. The company's competitors focus mostly on the Americas and EMEA, enabling it to expand in an underserved region. Japanese and Korean customers are expected to drive demand and adoption in this region to remain competitive against Chinese manufacturers.

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Commercial Model

GO-TO-MARKET

Key Decision Makers (KDM):

- **Semiconductors & Electronics**: R&D and Materials Heads are KDMs. Material scientists, AI & Data scientists, and formulation staff are part of the technical evaluation.
- **Chemicals, Energy, & Others:** R&D Heads are KDMs. Heads of Procurement are responsible for the technical and commercial assessment.
- **MedTech & Pharma:** Operations and new product development Heads are KDMs.

EVP, SVP, and C-level executives approve tools like CuspAI's with large annual contract values and have a career risk on the purchase, shaping how conservatively committees evaluate new providers.

Strategy

CuspAI focuses on acquiring customers by engaging directly with end-users of potential customers. It typically acquires customers by offering milestone-based contracts and initial projects for Proof of Concept (POC) prior to long-term commercial agreements. The company also deploys sales personnel to cover chemical industry events like GPCA and ADIPEC, which executives from large chemical and energy companies attend. It attempts to improve brand reputation and generate leads for customer acquisition in these events.

Sales teams also often pitch engagements to executives and engage technical experts with training and demonstrations, as purchasing requires support at all levels. The sales team briefs prospective customers on CuspAI's offerings and responds to inbound requests. The company offers specific use cases with measurable ROI and proof of value in short timeframes to onboard customers, starting in a specific area before expanding across use cases.

Sales Cycle

- The sales cycle length is typically six to nine months from first contact to pilot contract signing.
- Implementation typically takes three to six months.

Partnerships

CuspAI has multiple partnerships with large global companies, notably through its 'AI Materials Foundry' program. The program positions members as collaborators on materials projects and gives CuspAI visibility across industrial markets. Consortia like this are common among AI-powered material discovery providers as membership has no cost and preserves optionality for participants.

It also maintains technology partnerships with NVIDIA and Meta, facilitating market penetration and access to compute capacity. NVIDIA invests in many providers in the market including CuspAI, Periodic Labs, and Orbital Industries, which limits CuspAI's competitive advantage from these partnerships. NVIDIA refers customers to these providers to secure future compute demand and lowers computing costs for them.

The partnerships enable CuspAI to improve its credibility with industrial companies in the short term rather than enabling it to access customers directly, as most members are not paying customers. Market enthusiasm generated by partner announcements enables CuspAI to have higher pricing power when negotiating initial contracts with customers. However, it can only maintain long-term pricing power through successful POCs and sustained service levels over time.

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MONETISATION

PROFITABILITY

Pricing

CuspAI calculates a bespoke quote for each engagement rather than using a standard price list, given the nascency of the company.

- It prices custom programs upfront based on customer size, scope of the project, and expected usage.
- Platform access pricing is typically calculated based on a fixed subscription component and usage-based components tied to user licenses, compute consumption, and the scope of requested applications.
- It also quotes contract values upfront before execution for platform access and simulations.

The company offers 30% to 40% discounts for contract lengths above three years to secure future revenue. Pricing in the market remains inconsistent due to its nascency, and CuspAI is still developing its pricing strategy. Individual deals could close at more than 2x the value of comparable engagements when buyer enthusiasm is high.

For example, a basic platform access combined with custom projects for a large petrochemical customer is priced at approximately £750k per year. This includes usage focused on one or two applications like corrosion-resistant steels and catalysts to accelerate chemical processes. From 2026 to 2029, it could charge ACVs of around £3m as customers cover more material categories and require external lab synthesis and testing.

Costs

CuspAI builds its LLMs in-house, requiring large software development staff overheads and CapEx to develop and update its platform.

The majority of core operating costs are split across:

- **Labor contributes more than 70% of operating costs.**
- Software development personnel contribute approximately 50% of labor costs.
- Sales, administrative, and scientific staff contribute approximately 50% of labor costs.
- **Compute contributes more than 25% of operating costs.**

Scientific staff costs are expected to increase in line with custom-program growth given each custom program requires scientific staff. The remaining costs are expected to grow at less than 5% annually between 2026 and 2029, mostly driven by inflation. Scientific staff cost is expected to increase according to the growth in custom programs. Compute usage optimization is expected to limit further cost increases as the company refines its model training. Its UK-based operations support lower scientific labor costs than US-HQ'd competitors.

Margins

Gross Margins:

- Platform access has gross margins of 85% to 90%.
- Custom programs have gross margins of 5% to 15%.

Custom programs require dedicated scientific staff for each customer, driving increased costs and lower margins for this offering. Software platform access has low incremental delivery costs after development, driving higher margins.

EBITDA Margins:

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CuspAI had negative EBITDA margins in 2025 due to large development costs and commercial contracts remaining small. EBITDA Margins are expected to be 20% to 25% between 2033 and 2036.

Future Profitability Drivers:

The company is expected to drive higher profitability through migrating its consulting customers towards platform access and simulations, requiring less staff. As platform revenue scales without requiring additional scientific staff while revenue from custom engagements and consulting grows almost linearly with staff costs. The company requires successful pilot engagements to secure recurring subscriptions, and the speed at which CuspAI converts pilots into recurring subscriptions will drive profitability. The company must also sustain initial pricing power to achieve expected margins.

CONTRACTS

Terms

CuspAI contracts are typically structured in three stages, with milestone payments linked to each stage.

- POC contracts are used to demonstrate technical feasibility and provide value without requiring large upfront commitments from users.
- Pilot engagements are often 6 to 12 months long and have fixed fees depending on the size of the customer and the project scope.
- Full execution contracts typically run one to two years to enable customers to exit if returns are not realized.

Stage gates exist to align and meet customer expectations, driving customer retention rather than facilitating churn. Course corrections happen at each gate before problems develop, supporting customer confidence and increasing conversion to full execution contracts.

Commercial terms:

- Contracts are scoped and priced fully upfront.
- Contracts include options and add-ons to cover wider material families and expand scope.
- Contracts typically lock pricing for one to two years.
- Co-development projects include terms covering IP allocation, contract length, and usage rights.

Royalties:

Customers are subject to recurring payments tied to the revenue generated by materials created with CuspAI's tool. This creates an additional revenue stream if the discovery reaches production. Royalties are expected to generate the majority of CuspAI's revenue from customers with successful applications of materials discovered. Royalties vary by product line.

- Platform access has royalty rates of approximately 1% on materials discovered through the platform.
- Custom programs have royalty rates of 3% to 5% on materials discovered through end-to-end engagements.
- Customers can negotiate one-time lump sum fees with milestone payments instead of ongoing royalties.

Retention

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Strategy: CuspAI assigns a dedicated customer success team and account lead to each commercial account, with a point of contact tracking usage and ROI. The level of support depends on contract value and whether it involves co-development or basic platform access. Retention is higher in custom material discovery programs due to higher integration into customers' systems.

Churn Drivers:

Churn concentrates at the pilot conversion point, with only 40% to 50% of pilot customers transitioning to full execution engagements. Pilot churn typically occurs when models generate valid candidates that fail to become revenue-generating materials, preventing CuspAI from demonstrating ROI. Low-quality and unstructured customer data also limits value generation and drives churn.

Competitive takeaway drives churn in specific applications like drug discovery within life science customer segments, as providers with specialized models often deliver higher volume or quality material candidates at a lower cost. Royalties also drive churn as customers bound to pay a recurring royalty on discovered materials have an incentive to end contracts just before discovery and recreate material in-house using the progress achieved with CuspAI's tools.

Barriers to Churn: The company has an annual retention rate of 80% to 85% in full execution contracts, due to high switching barriers after full implementation. Customers replacing CuspAI require three to six months of change management, including retraining staff and rebuilding workflows around a new platform. KDMs who approve contracts with ACVs above £1m have responsibility and a career risk on the purchase, which motivates them to avoid churning and to justify the decision even with low or negative ROIs. Retention could reach up to 90% for strategic accounts as CuspAI integrates its platform more deeply into customer systems.

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Competitors

MARKETS

SUB-MARKETS

Generative Materials Discovery	Providers offer R&D custom programs that generate and validate candidate materials against target properties. This includes end-to-end discovery of new materials delivered through custom R&D programs.
Materials Informatics Platforms	Software platforms that organize, model, and optimize material data, which customers deploy internally to run their own discovery programs.
Autonomous Experimentation	Robotic laboratories that integrate generative design engines with automated synthesis and testing hardware, manufacturing and patenting materials in-house.

CUSPAI LTD.

FTEs	51 to 200 (2026)
Revenue	£7m - £12m (2025)
Ownership	VC backed (Series B)
Overlap	Pure player
Markets	Generative Materials Discovery ↑ · Materials Informatics Platforms – · Autonomous Experimentation –

Market Position: CuspAI has one of the widest material coverage among AI discovery providers, targeting mostly semiconductors, advanced manufacturing, and laboratory data ecosystems.

Technology Architecture & Synthesis: CuspAI executes DFT at a large scale to train its AI models on the results so the models can make predictions of how atoms interact in materials faster than running the underlying simulations. It also develops its own models instead of relying on third-party LLMs. This enables the company to offer scientifically grounded models that enable higher credibility than core competitors by focusing on digital R&D software rather than legacy system integration and physical labs.

Commercial Model: CuspAI focuses on custom R&D programs that deploy computational chemists to address customer skill gaps, generating most of its revenue from these engagements. IP requirements and royalties often limit customer acquisition as the company requires rights to train models on customer testing data, which some customers refuse due to data privacy concerns.

Specialization and Depth: CuspAI targets semiconductor customers with its specialized model, but its model also aligns with secondary customer use cases found in chemical and sustainability applications. It specializes in porous materials, sorbents, and catalysts for industrial applications, and in polymers, coatings, and soft materials in MedTech applications. It does not specialize in process-driven applications and metals, but covers some metals.

ORBITAL INDUSTRIES

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FTEs	11 to 50 (2025)
Ownership	VC backed (Series B)
Overlap	Pure player
Markets	Generative Materials Discovery ↑ · Materials Informatics Platforms ↑ · Autonomous Experimentation ↑

Market Position: Orbital Industries (previously Orbital Materials) is the closest competitor to CuspAI, particularly in O&G evaluations for sorbents, porous materials, and clean tech catalysts. It often wins customers against CuspAI who require providers offering a full-stack solution in-house.

Technology Architecture & Synthesis: its Lumis model is tailored to materials and trained on large and varied material datasets. The company targets metal-organic frameworks and zeolites, which are porous materials with a high surface area used for carbon capture and gas separation. Orbital Industries integrates digital generative design with physical lab automation, automated synthesis hardware, and robotic testing to manufacture and patent proprietary materials in-house. Its wider technology coverage enables it to sometimes win competitive tenders against CuspAI for customers who prefer contracting a single provider.

Commercial Model: it develops and patents its own materials instead of selling software. This closed-loop approach captures IP value from discovered materials but requires capital for synthesis and testing capacity that software-led providers avoid.

Specialization and Depth: Orbital Industries targets CO2 applications, water treatment, clean tech, and catalysts, and serves sorbent and catalyst producers as its ideal customers. Its material specialization overlaps with CuspAI in carbon capture and refining catalysts, but it lacks semiconductor applications.

CITRINE INFORMATICS

FTEs	51 to 200 (2026)
Revenue	£9m - £12m (2025)
Ownership	VC backed (Series C)
Overlap	Pure player
Markets	Generative Materials Discovery — · Materials Informatics Platforms ↑ · Autonomous Experimentation —

Market Position: Citrine Informatics is a legacy provider in the material discovery market. It has a wide material coverage and competes against CuspAI for large enterprise customers.

Technology Architecture and Synthesis: it offers an enterprise informatics platform that organizes and exploits heterogeneous materials and process data. Customers deploy its software internally and connect it to legacy R&D systems across structured and unstructured datasets, which gives it a competitive advantage over CuspAI. It often wins against CuspAI in tenders where customers prioritize material data infrastructure.

Commercial Model: It sells enterprise software subscriptions, and customers run discovery programs with their own teams on top of the platform. This delivery model enables it to scale with limited incremental costs and to avoid the IP friction that provider-delivered discovery programs create for customers.

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Specialization and Depth: it covers formulations, specialty chemicals, coatings, adhesives, and polymer applications. It does not deliver end-to-end discovery programs, driving customers to contract Citrine Informatics and CuspAI together when they require full outsourced R&D capacity.

PERIODIC LABS

FTEs	11 to 50 (2026)
Ownership	VC backed (Seed)
Overlap	Pure player
Markets	Generative Materials Discovery ↑ · Materials Informatics Platforms ↑ · Autonomous Experimentation ↑

Market Position: Periodic Labs was launched in Q3 2025, and it is still developing its commercial offering.

Technology Architecture and Synthesis: it operates a robotic laboratory that runs autonomous experiments, integrating generative design with physical synthesis and testing. Periodic Labs provides more advanced integration capabilities than CuspAI, and it develops and patents its own materials. The company is building scientific capability beyond materials, including weather forecasting models.

Commercial Model: its robotic laboratory model monetizes discoveries as patented materials instead of software subscriptions. Its public targets like developing room-temperature superconductors have a high risk profile as commercial outcomes depend on scientific breakthroughs that can take more than 10 years to materialize. It often loses customers to CuspAI who require commercially available capacity within five years.

Specialization and Depth: It specializes in semiconductor and electronics materials like thermal interface materials and dielectrics. Periodic Labs does not specialize in applications to serve petrochemical and refining customers.

ADJACENT COMPETITORS

Lila Sciences	Generative Materials... ↑ · Autonomous Experimen... ↑	Lila Sciences runs a robotic laboratory conducting materials research across catalysts and high-value molecules. The company delivers high-risk contract R&D engagements on a royalty basis rather than selling software. It focuses on life sciences applications for pharma customers, covering drugs and fermentation. The company often wins against CuspAI in tenders for large pharmaceutical customers. It also offers material applications for industrial and chemical customers, but often loses tenders for semiconductors and chemical customers against CuspAI.
Radical AI	Generative Materials... ↑	Radical AI has a similar technology approach to CuspAI. It runs DFT at large scale and trains ML models on the outputs, competing directly with CuspAI in generative materials discovery. It has less financial backing than the leading providers, which limits its ability to compete and innovate as fast as CuspAI.

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SandboxAQ	Generative Materials... ↑	SandboxAQ has a wide application coverage including drug discovery, materials, cybersecurity, and wider applications of specialized LLMs. The company started as a division of Alphabet and has been independent from the group since 2022. It has a similar technology approach to CuspAI, focusing on different customer segments including government and pharmaceutical companies.
Aionics	Generative Materials... ↑	Aionics competes against CuspAI in AI-powered materials discovery, using high-performance computing to discover and optimize materials across chemicals, semiconductors, and battery storage. It targets applications like chip cooling liquids.
Noble AI	Materials Informatic... ↑	NobleAI competes against CuspAI by providing material discovery software. It specializes in regulatory compliance and sustainability, targeting industrial and O&G companies. The company represents a moderate threat for CuspAI in early-stage pilots where customers test multiple vendors in parallel before selecting one.

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Outlook

GROWTH DRIVERS

DRIVER

Semiconductor customers are raising R&D budgets

REVENUE IMPACT

↑

COMMENTARY

Chip demand growth is supporting higher earnings for semiconductor customers, directing larger budgets toward discovery tools that support chip performance chemistry. CuspAI's models cover the material classes prioritized by these customers, making budget growth convert into engagements without new model development. This is driving increased revenue per account and customer acquisition in this segment.

DRIVER

Energy producers are industrializing AI workflows

REVENUE IMPACT

↑

COMMENTARY

Petrochemical companies are moving from isolated pilots into integrated workflows spanning exploration, production, and refining operations. CuspAI benefits as customers increase spending with AI-driven material discovery providers, with each added use case increasing platform usage. This is driving growth in EMEA and supporting the transition from pilots to multi-year engagements.

DRIVER

Environmental regulation is driving adoption

REVENUE IMPACT

↑

COMMENTARY

PFAS restrictions, rules on chlorinated compounds, and national sustainability commitments are driving chemical producers to discover compliant replacement materials. Regulatory pressure is expected to continue driving adoption of material discovery tools. New environmental regulation expands the addressable market that requires discovery tools and CuspAI could capture this demand as its models cover sorbents and remediation chemistries. Regulation around material discovery also raises entry barriers for new entrants, protecting the market shares of providers like CuspAI.

DRIVER

Government funding is sustaining customer budgets

REVENUE IMPACT

↑

COMMENTARY

Government grants to technology and materials companies are increasing capital in customer R&D budgets. Semiconductor and energy companies receive larger grants, contributing to budgets in CuspAI's target customer segments. Grant deployments are expected to continue, supporting customer budgets in the long-term and driving sustained spend until 2030.

DRIVER

REVENUE IMPACT

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Competitive development is reducing the technical gap

COMMENTARY

CuspAI's modeling approach has become a common industry practice, and the technical lead it held at founding is narrowing as competitors run similar DFT and machine learning technologies at scale. This could slow customer acquisition if customers find equivalent technical capabilities at lower prices. It could also further reduce margins in the custom programs, which are supported by high pricing power.

OPPORTUNITIES AND RISKS

THEME

OUTLOOK

The company could migrate customers from consulting to platform delivery

COMMENTARY

CuspAI could transition its custom program customers to platform access as the software matures, converting labor-bound revenue into scalable software revenue to increase EBITDA margins. The company's revenue mix could shift toward the platform as customers with computational chemistry talent adopt the software first, enabling wider commercial adoption as the technology matures.

THEME

OUTLOOK

CuspAI could expand in Korea and Japan

COMMENTARY

Korean and Japanese battery and electronics producers require discovery solutions to compete against Chinese competition, representing an expansion opportunity for CuspAI. Long sales cycles discourage competing providers from targeting Japan and Korea, leaving large whitespace for the company to capture. Battery and electronics manufacturers in this region require applications within CuspAI's material coverage, and the company could grow faster in APAC than in other regions if it acquires one or two large conglomerate customers.

THEME

OUTLOOK

CuspAI could target mid-market customers

COMMENTARY

CuspAI could target mid-sized companies with leaner model versions and channel partnerships, opening a customer pool that premium pricing models leave unserved. Early adoption in this segment could increase growth rates compared to focusing solely on enterprise customers. This requires simpler product packaging, and conversion depends on the company prioritizing customer acquisition over pricing.

THEME

OUTLOOK

Data privacy objections could limit model improvement

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COMMENTARY

If customers refuse to share testing and R&D data with CuspAI, its models could lose training data. Chemical customers evaluating the company often have data privacy concerns as they attempt to prevent competitor leakage. Model development could slow while competitors with wider data access improve, threatening CuspAI's technology advantage.

THEME

Compute and energy limitations could further delay model improvements

OUTLOOK



COMMENTARY

CuspAI has limited access to compute and energy as it does not control infrastructure capacity, limiting its model training. If compute and energy costs rise without corresponding revenue growth, CuspAI's margins could decline.

CuspAI must maintain partnerships to improve model training as they provide preferential compute access. This further enables the company's platform delivery, increasing its ability to serve customers at scale.

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